

Assignment:

Design a Smart Traffic Light System

Objective:

Apply **Decomposition, Pattern Recognition, Abstraction, and Algorithm Design** to design how a smart traffic light should operate at a busy 4-way junction.

Scenario

You have been asked to design the logic behind a **Smart Traffic Light System** that improves traffic flow, reduces waiting time, and prevents accidents. The system must handle:

- Peak hour traffic
- Pedestrian crossings
- Emergency vehicles
- Sensor-based vehicle detection
- Night-time low-traffic mode

Your task: Use the **four pillars of computational thinking** to plan how this system works.

Assignment Instructions

1. Decomposition (Break the problem down)

Identify **6–10 sub-tasks** required to design the smart traffic light system. Examples: detecting cars, controlling timers, switching signals, etc.

2. Pattern Recognition (Identify trends and behaviours)

List **at least 5 patterns** the system may observe, such as:

- rush-hour traffic surges
- pedestrian behaviour
- emergency vehicle siren detection
- night-time low traffic patterns
- vehicles bunching at certain lanes

3. Abstraction (Filter what matters vs. what doesn't)

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Identify:

- **5 essential variables** the system must monitor
- **5 irrelevant variables** it should ignore

Essential vs. irrelevant details help simplify the system.

4. Algorithm Design (Create step-by-step logic)

Write a **clear 10–12 step algorithm** describing exactly how the traffic light decides:

- which light turns green
- how long each light stays green
- when to prioritize pedestrians
- when to override signals for emergency vehicles

This should be written in plain English and *not code*.

5. IPO Model (Mapping the System)

Map the system using the **Input-Process-Output** pattern:

- **Input:** List the data or materials entering the system (e.g., sensor data, pedestrian button state, emergency signals).
- **Process:** Summarize the core transformation logic (e.g., evaluating density, checking timers).
- **Output:** Define the final result (e.g., specific signal changes, display of pedestrian "Walk" sign).

6. Logic Flowchart (Visual Representation)

Create a visual diagram of your algorithm using **standardized graphical symbols**:

- **Ovals:** Mark the **Start** and **End** of the logic.
- **Parallelograms:** Represent **Inputs** (reading sensors) and **Outputs** (changing light colors).
- **Rectangles:** Represent a specific **Process** (e.g., "Set timer to 30 seconds").

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- **Diamonds:** Represent **Decisions** (e.g., "Is an emergency vehicle detected?"). You must include clearly labeled **True/False** paths.
- **Flowlines:** Use arrows to show the direction of the system's control

Submission Format

Learners should submit their work in the following format:

- A PDF document
- A single page summary of the proposed solution

Evaluation Criteria

Criteria	Weightage	Description
Technical Application of Symbols	40%	Correct use of Ovals, Parallelograms, Rectangles, and Diamonds in the flowchart.
Completeness of System Logic	25%	Ensuring all scenarios (Emergency, Peak Hour) are represented without broken paths.
Accuracy of Decision Logic	20%	Effective use of Diamond symbols to demonstrate a functioning feedback loop or override logic.
Realism and Practicality	15%	Whether the step-by-step logic and IPO mapping are practical for a real-world 4-way junction